TEST CODE **02107032**

FORM TP 2012145

MAY/JUNE 2012

C A R I B B E A N   E X A M I N A T I O N S   C O U N C I L

ADVANCED PROFICIENCY EXAMINATION

BIOLOGY

UNIT 1 – Paper 032

ALTERNATIVE TO SCHOOL-BASED ASSESSMENT

2 hours

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of THREE questions. Answer ALL questions.
2. You are advised to take some time to read through the paper and plan your answers.
3. Write your answers in the spaces provided in this booklet.
4. You may use a silent, non-programmable calculator to answer items.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

1. Hydrogen peroxide (H_2O_2) is a poisonous by-product of metabolism that can damage cells if it is not removed. Catalase is an enzyme that speeds up the breakdown of hydrogen peroxide into water (H_2O) and oxygen gas (O_2). Catalase is found in plant and animal tissues and is especially abundant in the cells of potato and liver.

You are required to conduct a simple investigation into the **effect of substrate concentration on the rate of reaction of catalase**. Carefully read the instructions before starting the experiment.

The source of catalase is potato tissue. The indication of enzyme activity in this experiment is the height of the column of oxygen bubbles produced. See Figure 1 below.

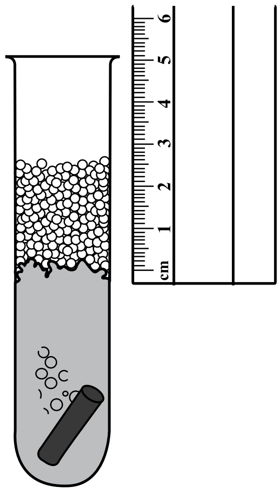

Figure 1. Demonstration of measurement of column of bubbles

(a) __**Procedure**__

(i) You are provided with eight test tubes labelled A, A1, B, B1, C, C1, D and D1, each containing 10 cm^3 of solutions as follows:

Test tubes A, A1	5% hydrogen peroxide
Test tubes B, B1	10% hydrogen peroxide
Test tubes C, C1	20% hydrogen peroxide
Test tubes D, D1	distilled water

Caution: Hydrogen peroxide is corrosive. You must use safety glasses and gloves when handing the test tubes.

(ii) You are provided with eight cylinders of potato, each 2 cm × 0.5 cm, in a Petri dish.

(iii) Using a forceps, add one potato cylinder to Test Tube A and observe for release of bubbles for 3 minutes.

(iv) Measure the height (mm) of the column of bubbles in the test tube using a ruler, as shown in Figure 1.

(v) Record your result for Test Tube A in Table 1.

(vi) Repeat the procedure as outlined in Steps (iii), (iv) and (v) for the remaining solutions in Test Tubes A1, B, B1, C, C1, D and D1.

[8 marks]

TABLE 1.

Test Tube	**Height of Bubbles (mm)**	**Average Height of Bubbles (mm)**
A **5% H_2O_2**		
A1 **5% H_2O_2**		
B **10% H_2O_2**		
B1 **10% H_2O_2**		
C **20% H_2O_2**		
C1 **20% H_2O_2**		
D **Distilled water**		
D1 **Distilled water**		

(b) Write an appropriate title for Table 1.

____

____

[1 mark]

(c) Calculate the average height of the bubbles for EACH concentration of hydrogen peroxide and for the distilled water. Record your results in Table 1 on page 3. **[2 marks]**

(d) Based on the results recorded, what can be deduced about the relationship between substrate concentration and enzyme activity?

[2 marks]

(e) State the purpose of the distilled water in Test Tubes D and D1.

[1 mark]

(f) Outline how the procedure used in this experiment could be modified to investigate the effect of **temperature** on enzyme activity.

[2 marks]

Total 16 marks

2. (a) You have been provided with plastic ties of three different colours and three different lengths.

(i) Use the plastic ties to construct a model which shows the structure of **bivalents for two chromosomes** of different lengths at the end of the **prophase stage of Meiosis I**.

For EACH chromosome use two different colours of the same lengths to represent the parts of the bivalent derived from each member of a pair of homologous chromosomes. Use the third colour to represent centromeres. **[4 marks]**

(ii) Show ONE **chiasma** on your model. **[1 mark]**

(iii) In the box below, **arrange your bivalents as you would expect to see them at the end of the prophase stage of Meiosis I.**

Draw an outline of the arrangement in the box.

Use the transparent tape provided to attach your bivalents to the paper in the box below. **[2 marks]**

(iv) In box above, draw the centrioles and spindle fibres in relation to your bivalents, as they would be seen at the end of Prophase I. **[2 marks]**

GO ON TO THE NEXT PAGE

(b) Colour blindness is inherited as a sex-linked recessive disease with the mutant gene carried on the X chromosome.

A colour-blind male marries a female who is heterozygous for colour blindness.

(i) State the genotypes of the parents.

Male: ______________________________

Female: ______________________________

[2 marks]

(ii) In the space provided below, draw a Punnett square to show the possible offspring.

[3 marks]

(iii) State the phenotypes of the male and female offspring with respect to colour blindness.

Males: ______________________________

____

Females: ______________________________

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[2 marks]

Total 16 marks

3. (a) Specimen A is a stained longitudinal section of the testis of a mammal. Examine the specimen using the lower power of a compound microscope.

(i) In the box below, make a labelled plan drawing to show the distribution of the tissues in the testis. **[7 marks]**

(ii) For EACH type of tissue labelled, state its function next to the corresponding labels on your drawing. **[2 marks]**

(b) Figure 2 is a diagram of the human female reproductive system.

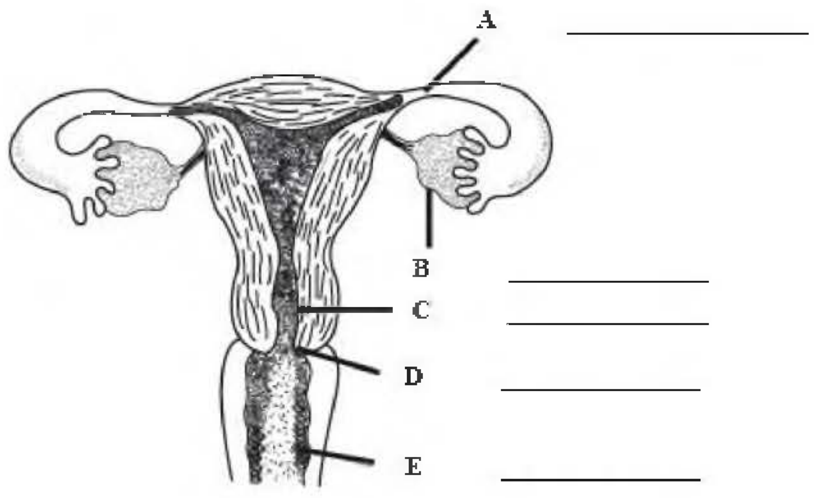

Figure 2. Human female reproductive system

Source:http://thornlea.sharpschool.com/UserFiles/Servers/Server_119514/File/Department%20 Document%20%28BY%20SUBJECT%29/Documents%20Physical%20 Education%20Department/Cassie%27s%20Folder/Gr%209/Female%20 Reproductive%20System%20-%20Internal.jpg

(i) On the diagram in Figure 2, identify the structures labelled A, B, C, D and E. **[5 marks]**

(ii) On the diagram in Figure 2, indicate, using annotations, where fertilization of the ovum and implantation of the zygote occur. **[2 marks]**

Total 16 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.